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OPERATING INSTRUCTIONS

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SGS-CSTC STANDARDS TECHNICAL SERVICES CO.,LTD IND016
2024/1/18

Equipment Name	Oil/Gas Separator
Dwg No.	20980145-083-001
Item No.	P-2362-001-025
Prepared by	[illegible]
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Premise: Informations given on the nameplate and on the Technical Documentation, supplied together these Operating Instructions, are part of these Operating Instruction and not necessarily repeated here.

0 BASIC INFORMATION OF THE EQUIPMENT

Design Pressure PS	1.44Mpa	Fluid	Left Cavity: Air， Right Cavity:Oil&Air
Design Temperature TSmax/min	150/0℃	Fluid property	Group2
Test pressure TS (Horizontal or vertical)	2.06Mpa	Volume	130L/23L
RT percentage	N/A	Filling ratio	N/A
PWHT	N/A	PED category	III
Corrosion allowance	1.0mm	Empty weight	N/A
Safety devices	N/A	Test weight	N/A
Fatigue life	N/A		

The following selected loads are considered during the design of this equipment. The loads unselected are subject to user's consideration before putting this equipment into use.

- ■ Internal Pressure
- □ External Pressure
- ■ Weight Of The Vessel
- □ Weight Of Normal Condition Under Operating Condition
- ■ Weight Of Normal Condition Under Test Condition
- □ Superimposed Static Reactions From Weight Of Attached Equipment
- □ The Attachments Of Internals
- ■ The Attachments Of Vessel Supports(Skirt, Legs Saddles Etc.)
- ■ The Attachments Of Lifting Lugs
- □ Cyclic And Dynamic Reactions Due To Pressure
- □ Cyclic And Dynamic Reactions Due To Thermal Variations
- □ Cyclic And Dynamic Reactions Due To Equipment Mounted On The Vessel
- □ Cyclic And Dynamic Reactions Due To Mechanical Loads
- □ Wind Reactions
- □ Snow Reactions
- □ Impact Reactions Such As Those Due To Fluid Shock
- □ Temperature Gradients
- □ Differential Thermal Expansions
- □ Abnormal Pressure
- ■ Test Pressure And Concident Static Head Acting During The Test

1 PROTECTION FOR TRANSPORT, STORAGE, HANDLING

'1.1 Transport and storage

The equipment has been protected for the transport and storage prior to erection as necessary against physical damage and / or corrosion which may occur due to the method and conditions of transport and the time which may elapse before erection, up to maximum of 3 months. For longer periods, protection shall be restored by user at storage site, with particular attention to the protection of machined surfaces, and the compatibility therewith, as regards corrosion, of any material used for physical protection.

'1.2 Handling

Only for equipments provided with brackets / lugs: during loading and unloading operations brackets / lugs shall be used with purpose to avoid the possibility of distortion of the equipment and any of its parts.
It is strictly forbidden to lift or to move the equipment by fastening others braces like nozzles or supports without written permission of the manufacturer. Any damage must be notified immediately to the manufacturer.

2 MOUNTING, INCLUDING ASSEMBLING

‘2.1 Installation of the equipment

The equipment shall be installed in accordance with applicable requirements of State or Local Authorities, and with applicable regulations.
Before installing the equipment, personnel must check the integrity of the nameplate and all the information are well readable.
Personnel must control than the equipment is in good conditions and no damage has been occurred during transport and handling.
The position of the equipment is indicated on the drawing(s). If provided with supports, the equipments must be installed by the supports forecast on the equipment.
It is not allowed to support equipment by means of adjacent piping, unless so specifically stated by the designer.

‘2.2 Levels

The equipment shall be installed to the correct levels and vertical lines within the limits agreed with the owner/user (see drawing(s)).
To facilitate drainage of static horizontal equipment, the longitudinal fall should be at least 1 in 200.

‘2.3 Foundations

Foundations shall be adequate to support the equipment and its maximum content.
Only for equipments provided with supports (brackets, lugs, legs, etc.): the type of equipment support (brackets, lugs, legs, etc.) shown in the assembly drawing(s) may be used to transmit loads to foundations. Each hole in the base plates must have its own foundation bolt. If not indicated on the drawing(s), the material for anchorage bolts shall be at least Class 8.8 of EN 20898-1 or equivalent, and bolt diameter shall be not smaller than the hole diameter minus 3 mm.

‘2.4 Equipment placement and access

The equipment shall be installed with sufficient clearance from associated structures and equipment to provide safe, efficient working by operators and to provide ready access for cleaning, inspections, and maintenance.
Equipments having manholes, handholes, or cover plates to permit inspection of the interior shall be so installed that these openings are accessible.
Only for equipments provided with supports: supports shall be so arranged as to provide adequate facility for the inspection of every part of the equipment, with sufficient access to all parts of the exterior to permit proper inspection, unless the equipment is of such size and is so connected that it may readily be removed from its permanent location for inspection.

Improper placement next to other equipment, chemicals, flammable materials, etc. may lead to a premature failure. Fire can weaken the equipment materials; chemicals may cause severe corrosion and failure before the next scheduled inspection; inadequate structural supports can lead to failure of nozzles or adjacent piping; vibration or failure of adjacent equipment may damage the equipment. These are just a few of the possibilities.

Where necessary, ladders and platforms shall be installed to give access to valves and other control device (if any) placed on equipment connections.

Safety and control devices (if any), shall be located and installed so that they are readily accessible for operation, inspection, maintenance and removal.

‘2.5 Assembly

The equipment has to be assembled complete with all its parts included blind flanges and internals (if any) fully assembled to the equipment.
The assembly of equipment to its adjacent connections (inlet, outlet, etc.) must be done by well trained personnel.
The type of connections may be (see drawing(s)):
Flanged connection
Threaded connections
Quick opening connections
Butt weld ends nozzles

The flanged connections must be ensured with gaskets and bolt indicated on the drawings if not already forecast on the supply.
For the threaded connections must be used special sealing to be chosen according to the operating temperature. In case of no indications before assembly the threaded connections, assembler must inform the manufacturer in order to supply specific information.
The assembly of quick special opening（if any）must be made with the bolts and accessories supplied with the equipment.
It is strictly forbidden to change or modify the bolts or the gaskets or any other accessories supply with the equipment to join the connection.

‘2.6 Filling and discharge
Where a discharge connection is provided to discharge the fluid content, the discharge piping shall be attached to a valve and shall lead to a safe valve location.
The filling and discharge system shall be designed provided and installed by User / Manufacturer of the assembly under their responsibility. Filling and discharge shall be made in a manner to prevent danger to person or damage to the operation of equipment or environment and preferably to a place where the discharge is visible.

Only heat exchangers: filling and discharge of heat exchanger shall be executed gradually with simultaneous circulation of fluids through both sides.

‘2.7 Substitution of removable parts
In general, removable parts as bolts, nuts, blind flanges, plugs, etc. may be replaced, if necessary (e. g. lost) with parts of same material standard specification, grade, size and (if applicable) pressure rating class indicated on the bill of material shown on the drawing(s) and other design documents.
Substitutions of removable parts may fall in the scope of in-service laws and regulations valid at place of installation, to be followed by the user.
In case of same type of material should be not available, the responsible of repair or substitution should notify the lack of fastening and he can not install others materials without previous authorization of the PED manufacturer.

‘2.8 Repairing
User is obliged to inform the manufacturer for any maintenance on the equipment that can seriously modify the safe operating condition of the equipment. Any repairing on the equipment must be carried out by qualified personnel.
Substitution of parts, or repair, by welding fall in the scope of in-service laws and regulations valid at place of installation, to be followed by the user.
The responsible of substitution of parts or repair by welding can not operate without previous authorization of the PED manufacturer. In lacking of this authorization, responsibility and warranty will be considered expired for the whole equipment in which such operation should have been made.

If not in conflict with local in-service laws and regulations, following should be used as code of practice:
‘- National Board Code ANSI/NB-23 latest edition.
‘- Only for vessel in petroleum refinery service, in addition, API RP 510 latest edition.

3 PUTTING INTO SERVICE

Only for equipments provided with supports (brackets, lugs, legs, etc.): the equipments must be fixed to its supports before operating.

Qualified personnel should check at least:
- Correct position of the equipment;
- No material has been left during assembling in the internal of the equipment;
- Cleanness of the internal according to the drawing(s);
- All the connections are fully and correct assembled;
- All the joint are correct fitted;
- All the bolts are correct closed;
- The foundation bolts (if any) are all in place and correct closed;
- All the safety instrument (if any) are in place and correct assembled;
- No evidence of damage to any parts of the equipment;
- Safety devices and alarms (if any) are connected on the whole of the equipment.

Starting the operating of the equipment (or of the assembly of which the equipment is part) must be done according the PED operating instructions and plant engineering procedure of the owner/user.

If the equipment is supplied with stickers, in order to inform about the risk of pressurization prior of installation of safety devices, they shall be considered and strictly followed.

All the information regarding the operating conditions (pressure and temperature) are indicated on the nameplate installed on the equipment, and/or on technical documentation furnished with the equipment.
Only for vessels: the position of the nameplate is indicated on the drawing(s).
Operating the equipment to proper design data indicated on the nameplate must always be done by a qualified personnel.

At least at half of the operating pressure a step must be done in order to check the following point by a qualified personnel:

- Lacks;
- Correct fastening;
- Distortion or damage;
- Safety accessories are well visible and readable.

4 USE

The use of the equipment should be carried out only by specialized personnel.
The use of the equipment is strictly obliged to the data indicated on the nameplate and on the drawing(s), the operating instructions and/or other technical documentation furnished with the equipment.
It is forbidden to use the equipment for other scope and other pressure and temperature different from the ones indicated on the nameplate and drawing.

It is strictly forbidden to open any closure, joints or other parts during operating.
It is strictly forbidden to tight bolts or thread during operating.
During operation only authorized personnel can stand near the equipment and only after having followed all the safety instructions issued by the plant safety responsible of the owner/user.

Only for heat exchangers: the safe operation of the vessel is assured only if both sides are feeded with fluids of type, composition, service pressure and temperature indicated on the drawing. The risk of lack of feeding of the colder fluid, drop of its pressure, and of failure of feeding pumps shall have been analyzed with one of methods addressed on Par. 8 of these instructions, and the risk reduced with use of adequate protection system, able to stop the warmer flow, activate the alarms, etc. designed by User / Manufacturer of assembly; in any case, the Manufacturer of the vessel does not assume responsibility of the correct operation of the protection system.

5 MAINTENANCE, INCLUDING CHECKS BY THE USER

Maintenance must always be done at room temperature and atmospheric pressure. It is strictly forbidden any maintenance to either the equipment, parts, components or instruments during the operating of the equipment.
The return to operating must be done after verifying that all the procedure indicated by the plant engineering are satisfied.
The equipments can reach its operating pressure and temperature indicated on the nameplate and drawing(s) and/or other technical documentation furnished with the equipment only after a positive check.

6 INSPECTION

Inspections should be made according to local laws and regulations for in-service and the Engineering/user plant procedure or, in lacking of any procedure, the National Board Code ANSI/NB-23 latest edition should be followed (if not in conflict with local laws and regulations for in-service).

Only for vessels in petroleum refinery service: in addition to above, API RP 510 latest edition shall be followed if not in conflict with local laws and regulations for in-service).

The type of inspection given to equipment shall take into consideration the condition of the equipment and the environment in which it operates. This inspection may be external or internal, and use a variety of nondestructive examination methods. The inspection method may be performed when the equipment is operating on-stream or depressurized, but shall provide the necessary information that the essential sections of the equipment are of a condition to continue to operate for the expected time interval. On-stream inspection, included while under pressure, may be used to satisfy inspection requirements provided the accuracy of the method can be demonstrated.

Inspection of environment in which the equipment operates includes:

- External painting or coating in order to established if there is any possibility further ambient corrosion;
- Good conditions of instruments and safety apparatuses (if any);
- Tight of foundation bolts (if any).

‘6.1 EXTERNAL INSPECTION

The purpose of an external inspection is to provide information regarding the overall condition of the equipment. In particular:
Insulation or other covering (if any). If it is found that external coverings such as insulation and corrosion resistant linings are in good condition and there is no reason to suspect any unsafe condition behind them, it is not necessary to remove them for inspection of the equipment. However, it may be advisable to remove small portions of the coverings in order to investigate their condition and the condition of the metal.
Evidence of leakage. Any leakage of gas, vapor or liquid shall be investigated. Leakage coming from behind insulation coverings, supports or settings, or evidence of past leakage shall be thoroughly investigated by removing any covering necessary until the source is established.
Structural attachments. The equipments mountings (if any) shall be checked for adequate allowance for expansion and contraction, such as provided by slotted bolt holes or unobstructed saddle mountings. Attachments of legs, saddles, skirts or other supports (if any) shall be examined for distortion or cracks at welds.
Equipment connections. Manholes, reinforcing plates, nozzles or other connections should be examined for cracks, deformation or other defects. Bolts and nuts should be checked for corrosion or defects. Weep holes in reinforcing plates shall remain open to provide visual evidence of leakage as well as to prevent pressure build up between the equipment and the reinforcing plate. Accessible flange faces shall be examined for distortion and to determine the condition of gasket-seating surfaces.
Abrasives. The surface of the equipment shall be checked for erosion.
Dents. Dents in an equipment are deformations caused by their coming in contact with a blunt object in such a way that the thickness of metal is not materially impaired. In some cases, a dent can be repaired by mechanically pushing out the deformation.
Distortion. If any distortion is suspected or observed, the overall dimensions of the equipment shall be checked to determine the extent and seriousness of the distortion.
Cuts or gouges. Cuts or gouges can cause high stress concentrations and decrease the wall thickness. Depending upon the extent of the defect, it may be necessary to repair the area by welding or patching. Blend grinding may be a useful method of eliminating some minor types of cuts or gouges.
Surface inspection. The surfaces of shell and heads shall be examined for possible cracks, blisters, bulges, and other evidence of deterioration, giving particular attention to the skirt and to support attachment and knuckle regions of the heads.
Weld joints. Welded joints and the adjacent heat affected zones shall be examined for cracks or other defects. Superficial NDE is recommended.

‘6.2 INTERNAL INSPECTION (FOR EQUIPMENTS WHERE APPLICABLE)

The following activities shall be performed to support the inspection:
Inspection plugs and covers shall be removed;
Equipment shall be sufficiently cleaned to allow for visual inspection of internal and external surfaces.

If the geometry of the equipment does not permit physical access, appropriate instruments (e.g. mirror, magnifier, borescope, fiberscope) shall be used.

A general visual inspection is the first step in making an internal inspection. All parts of the equipment shall be inspected for corrosion, erosion, blistering, deformation, cracking and laminations if applicable. In particular:
Equipment connections. Threaded connections shall be inspected to ensure that an adequate number of threads are engaged. All openings leading to any external fittings or controls shall be examined as thoroughly as possible to ensure they are free from obstructions.
Equipment closures (if any). Any closures, which are used frequently in the operation of an equipment, shall be checked by the Inspector for adequacy and wear. A check shall also be made for cracks at areas of high stress concentration**.**
Equipment internals (if any). Where equipment is equipped with removable internals, these internals need not be completely removed, provided assurance exists that deterioration in regions rendered inaccessible by the internals is not occurring to an extent that might constitute an hazard or to an extent beyond that found in more readily accessible parts of the equipment. If a preliminary inspection reveals unsafe conditions, steps shall be taken to remove or repair such parts so that a detailed inspection may be made.
Corrosion.(if any) The type of corrosion (pitted or uniform), its location and any obvious data shall be established. Data collected for equipment in similar service will aid in locating and analyzing corrosion in the equipment being inspected. Welded seams and nozzles and areas adjacent to welds are often subjected to accelerated corrosion.
Only for vessels: the liquid level lines, the bottom and the shell area adjacent to and opposite inlet nozzles are often locations of most severe corrosion.

The above indicated inspections are not limiting to any eventual further control that the qualified personnel could by themselves carried out in order to establish the remaining life of the equipment.

‘6.3 INSPECTION INTERVALS

New equipment is placed in service to operate under their conditions for a period of time determined by the service conditions. If the equipment is to remain in operation, the allowable conditions of service and length of time before the next inspection shall be based on local laws and regulations in account of the condition of the equipment as determined by the inspection.
The maximum period before the first internal inspection or a complete on-stream evaluation of equipment shall not exceed one-half of the estimated life or four years, whichever is less.
Each next period between internal inspections or a complete on-stream evaluation of equipment shall not exceed one-half of the estimated remaining life of the equipment or two years, whichever is less. Where the remaining operating life is less then two years, the inspection interval may be the full remaining safe operating life.

Lifetime of new equipment
The estimated lifetime of new equipment is 10 years. The user, under his exclusive responsibility, may decide to extend the life on the basis of the records of the results of periodic inspections.

‘6.4 EQUIPMENTS SUBJECTED TO FATIGUE (if applicable and considered during design)
Equipments with pressure variation less than 5% of the design pressure
They are not considered to be subject to fatigue.
Equipments with pressure variation larger than 5% of the design pressure
Except otherwise specified in the design technical documentation, equipment not specifically calculated to resist fatigue is intended for maximum fatigue lifetime as follows:
For equipments wall thicknesses calculated by formulas according to ASME code(s):
‘- 1000 cycles, in which the pressure variation exceeds 20% of the design pressure, for equipments not containing pad type nozzles and nonintegral attachments; but 400 cycles, in which the pressure variation exceeds 15% of the design pressure, for equipments containing pad type nozzles and nonintegral attachments; or,
For equipments calculated by formulas according to European codes:
‘- 500 cycles.

If a condition of fatigue is applicable, following provisions apply.

Periodic inspections
‘a) Each equipment for which the number of allowable load cycles N has been fixed shall undergo an internal inspection at the latest when half of the load cycles fixed has been reached.
As a result of the type of operation, the intervals between internal tests may be shorter in accordance with laws and regulations of the country of installation.
The user is obliged to record the number of load cycles arising by suitable means and, if necessary, to arrange for internal inspection.
‘b) If the operating conditions assumed in the calculation deviate in terms of a grater cyclic loading or if damage to the pressure-bearing wall is to be expected before the end of the inspection intervals owing to other operational influences, they shall be reduced in accordance with national regulations.
‘c) In-service inspections are of particular importance; they permit early detection of incipient damage. For this purpose, the internal inspections shall be supplemented by non-destructive tests on highly loaded locations. Surface cracks tests and ultrasonic tests are the methods to be considered here. For the examinations of readily accessible areas, the outside surface of the equipment can also be subjected to ultrasonic testing.
‘d) If no cracks are detected during an internal inspection, the next internal inspection shall be performed following a special agreement within the shortest interval resulting from the national regulations however at the latest. When half of the fixed number of load cycles has been reached. This also applies if the allowable number of load cycles is exceeded.
‘e) The inspection described in a) to d) for cyclic loading during operation may be waived if the component is designed to withstand an operational load cycle number of 2000000.

Consideration of special operating conditions
If corrosion-induced crack formation (fatigue crack corrosion, strain-induced crack corrosion) hydrogen-induced crack formation in compressed hydrogen is to be expected, it shall be taken into account that the fatigue strength value will not only fall considerably below the value without these influences, but that it may also cause fatigue cracks even after a very high number of load cycles (>10000000).

Measures to be taken when the design lifetime has been reached
‘a) If the allowable number of load cycles for a component or the allowable value for cumulative damage according to calculation has been reached, non-destructive tests according to above shall be performed as completely as possible at a certain number of highly loaded locations to be laid down with the relevant third party.
‘b) If no cracks are found in the tests conducted as described in a), continuous operation is permitted. The prerequisite for this is that no fatigue damage is found in the non-destructive tests conducted in the inspection intervals which correspond to 50% of the operating time specified in a). After this operating time has been reached, further measures shall be agreed for each individual case in accordance with national regulations.

‘c) Should cracks or crack-like defects or other more extensive damage established in the tests carried out as described in a) or b), the component or the structural element concerned shall be replaced, unless continued operation is deemed to be permitted by virtue of appropriate measures to be agreed in accordance with national regulations.
‘d) The following design, manufacturing or process-related measures can be considered appropriate to allow continued operation:
‘d1) Removal of cracks by grinding. If as a result of grinding too thin a wall thickness is obtained, repair welding shall only be carried out in cooperation with the manufacturer within the context of national regulations;
‘d2) Grinding the welds to remove all notches;
‘d3) Removal of restraints to expansion: e.g.: replacement of cracked rigid stiffeners by joints not restraining expansion;
‘d4) Change in mode of operation.

‘6.5) CORROSIVE SERVICE (if applicable)

For corrosive service, record of wall thickness checks shall be up-dated and filed at field site and will be available to inspector performing the next inspection with purpose to permit to know and review the history of the equipment.

Design corrosion rate
Additional metal thickness over and above that required for the initial operating conditions is provided at least equal to the expected corrosion loss (mm) during the estimated lifetime of new equipment.
The design corrosion rate (mm/year) for new equipment is the ratio of design corrosion thickness to the estimated lifetime of equipment or 10 years period, whichever is smaller.

Corrosion rate for service conditions changed
Corrosion rate in mm/year is for new equipment or equipment for which service condition are being changed, under the full responsibility of the owner, one of the following methods shall be employed to determine the probable rate of corrosion from which the remaining wall thickness, at the time of the next inspection, can be estimated:
The corrosion rate as established by data collected by the owner or user on equipment in the same or similar service.
If data for the same or similar equipment are not available the corrosion rate as estimated from the inspector’s knowledge and experience with equipment in similar service.
If the probable corrosion rate cannot be determined by either of the above methods, on stream thickness determination shall be made after approximately 1000 hours of service. Subsequent sets of thickness measurement shall be taken after additional similar intervals until the corrosion rate is established.

Remaining life calculation
Where the corrosion rate controls the life of the equipment, the remaining life shall be calculated as follows:

$$\text{Remaining life (years)} = \frac{\text{t(actual)-t(required)}}{\text{corrosion rate}}$$

7 CREEP SERVICE (if applicable)

Creep range of materials used is above the maximum allowable temperature TS at which the equipment may operate.

8 PROTECTIVE DEVICES AND OTHER FITTINGS (IF PROVIDED BY MANUFACTURER OF EQUIPMENT)

The equipment may be supplied with stickers in order to inform about the risk of pressurization prior of installation of safety devices.

Safety devices and alarms are not part of supply. In any case each equipment shall be provided with protective devices in accordance with the requirements of PED and Construction Code indicated on the drawing(s).
Protective devices shall be conform to the law requirements of the country where the equipment shall be installed and, where required, acceptable to the Inspecting Authority competent for the survey of the plant in-service.
When any such device is not provided by the equipment, the User / Manufacturer of the assembly shall be responsible for ensuring that it is supplied and fitted prior to placing the equipment into service. Design of protective devices, which are not provided by the manufacturer of the equipment, is full responsibility of the User / Manufacturer of the assembly.

Protective devices and fittings shall be:

- located and installed so that they are readily accessible for operation, inspection, maintenance and removal;
- arranged to afford maximum protection against accidental damage.

The equipment, marked CE having been subjected to a conformity assessment procedure, at site of installation will operate as an item of an assembly. The Manufacturer of the equipment does not control if the assembly, of which the equipment is part:
'- is assembled and installed by a manufacturer or by an user;
'- is , or not, marked CE as a whole, by a manufacturer;
'- is, or not, subjected to local laws and regulations for putting them into service.

Compliance with in-service local laws and regulations and, in case of plants with hazard due to particular dangerous products, compliance with Directive 96/82/EC (in Italy implemented with D.L. 334 of 1//08/99) are exclusive responsibility of the user.

'8.1 PRESSURE RELIEVING DEVICES

Each equipment (or each chamber of a multi-chamber equipment) shall be protected by one or more pressure-relieving devices that shall prevent the pressure from rising to more then 110 percent of the design pressure of the equipment.

Types of pressure relief devices to be designed, installed, used
Device designed to reclose and prevent the further flow of fluid after normal conditions have been restored. The valve automatically discharges compressible fluid, or liquid to atmosphere or to a reduced pressure system, so as to prevent a predetermined pressure being exceeded. It is activated by the static pressure upstream of the valve.
Pressurization of the equipment is forbidden in lacking of one of following types of pressure relief devices:
Safety valve: the (compressible) is discharged to atmosphere;
Relief valve: the (non-compressible) fluid, (i.e. liquid) is discharged to atmosphere or to a reduced pressure system.

Non-reclosing pressure relief devices
As alternative of pressure relief devices, non-reclosing pressure relief devices, designed to remain open after operation, or rupture disk devices may be used under full responsibility of the Customer / user.

'8.2 INSTALLATION

Interconnected equipments
Equipments (or chambers of equipments) which are connected together in a system by piping of adequate capacity, may be considered as one unit in determining the number and capacity of pressure-relief devices provided that no valve is fitted that could isolate any equipment from the relief devices unless the equipment is simultaneously opened to atmosphere.

Inlet connection
The connection between the relief devices and the equipment shall be as short as practicable, shall have bore area equal at least to the area of the relief devices inlet, and shall not reduce the discharge capacity of the relief devices below the capacity required for the equipment. .

'8.3 DISCHARGE FROM PRESSURE RELIEF DEVICES

Safe discharge to atmosphere
(Except lethal fluids and other special fluids considered in the relevant application code) fluid may be discharged from a static equipment to atmosphere provided that the discharge is outside of and away from buildings, preferably though a vertical pipe to a height of at least 2 m above the equipment or the building there the equipment is installed.
All relief devices shall be arranged so that the discharge does not impinge on the equipment, and will not prevent effective operation of the device.

Safe discharge to lower pressure systems
Discharge to lower pressure systems is permissible provided that the receiving system can accept the additional load without causing an unacceptable back pressure.

Discharge pipes
Discharge pipes from pressure-relief valve shall be sized so that, under maximum discharge conditions, the built-up of back pressure at the outlet of the valve as the result of the valve discharging does not reduce the relieving capacity of the valve below that required to protect the equipment.
The bore of the discharge pipe shall be not less than the bore of the outlet of the pressure-relief device.

Discharge piping shall run as directly as practicable to the point of final release.
Discharge piping shall be adequately and independently supported to prevent transmittal of forces due to the mass of the tube, discharge reaction and the thermal expansion strain
Forces acting on safety or relief valve should be kept to a minimum under all conditions of operation.
Discharge pipes shall be designed to facilitate drainage or shall be fitted with an open drain to prevent liquid from lodging on the discharge side of the device.

Common discharge pipes
Where it is not feasible to provide each pressure-relief device with a separate discharge pipe, a discharge pipe common to a number of devices on one or more equipment may be used. The size of common discharge pipe serving two or more pressure-relief devices that may reasonably be expected to discharge simultaneously, shall ensure that the required total discharge capacity can be achieved. The total pipe area should be at least equal the sum of their outlet areas, with due allowance for pressure drop in the downstream sections.

9 IDENTIFICATION

Identification of the equipment is made by its serial no. The serial no is stamped on the nameplate and on the equipment nearby the nameplate support in order to avoid change of the nameplate.
Information shown on the nameplate are considered part of these instructions.
Drawing(s) and other technical documentation supplied with the equipment are considered part of these operating instructions for all information not specifically covered by these instructions.